DP Biology – A3.2 (HL only)

# Classification and cladistics

- Which tools help us sort organisms into taxonomic groups?
- How does cladistics differ from traditional classification?

HL statements A3.2.1–9. Textbook pp. 125–137.
Image: a stoat. Steve Hillebrand, USFWS, public domain.



# Objectives for each class

| Class | Statements | By the end of the class, you can… |
|---|---|---|
| 1 | A3.2.1–3 | Explain why we classify, the problems with ranked taxa, and the value of classifying by ancestry |
| 2 | A3.2.4–7 | Explain clades and the molecular clock, and build and analyse cladograms from sequence data |
| 3 | A3.2.8–9 | Use cladistics to test classifications, and explain the three domains |

# Why classify?

A3.2.1



Arum maculatum: one species, dozens of common names. H. Zell, CC BY-SA 3.0.

Millions of species: grouping makes diversity **manageable**.

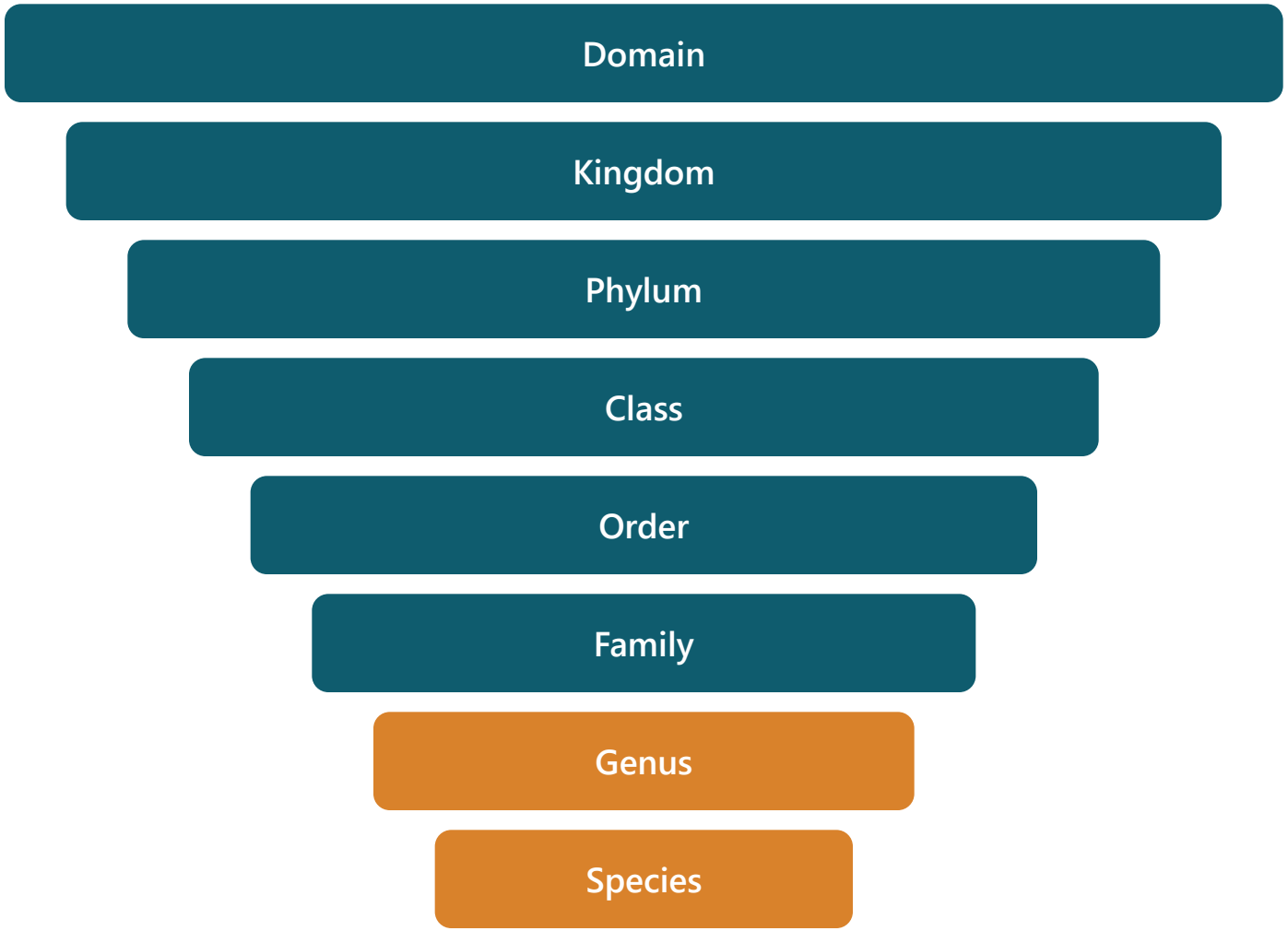
One **binomial** name works in every language, and a group lets us **predict** features of a new species.

### Try it

This plant has dozens of common names. Why is one scientific name better?

# The traditional hierarchy of taxa

A3.2.2




| Wolf | Daisy |
|---|---|
| **Animalia** | Plantae |
| **Chordata** | Angiospermophyta |
| **Mammalia** | Magnoliopsida |
| **Carnivora** | Asterales |
| **Canidae** | Asteraceae |
| **Canis** | Bellis |
| **C. lupus** | B. perennis |

# Problems with ranked taxa

A3.2.2

**No natural rule**

What counts as a family or order is arbitrary

**Unequal ranks**

One genus: 1 species; another: hundreds

**Convergence**

Look-alike groups mix unrelated organisms

### Reptiles without birds

Birds descend from reptile ancestors, yet traditional Reptilia leaves them out. The ranks do not match the branching pattern evolution actually produced.

# Classifying by evolutionary relationships

A3.2.3

Every member of a group descends from one **common ancestor** and shares features inherited from it.

So we can **predict** features of an unstudied member.

### Example

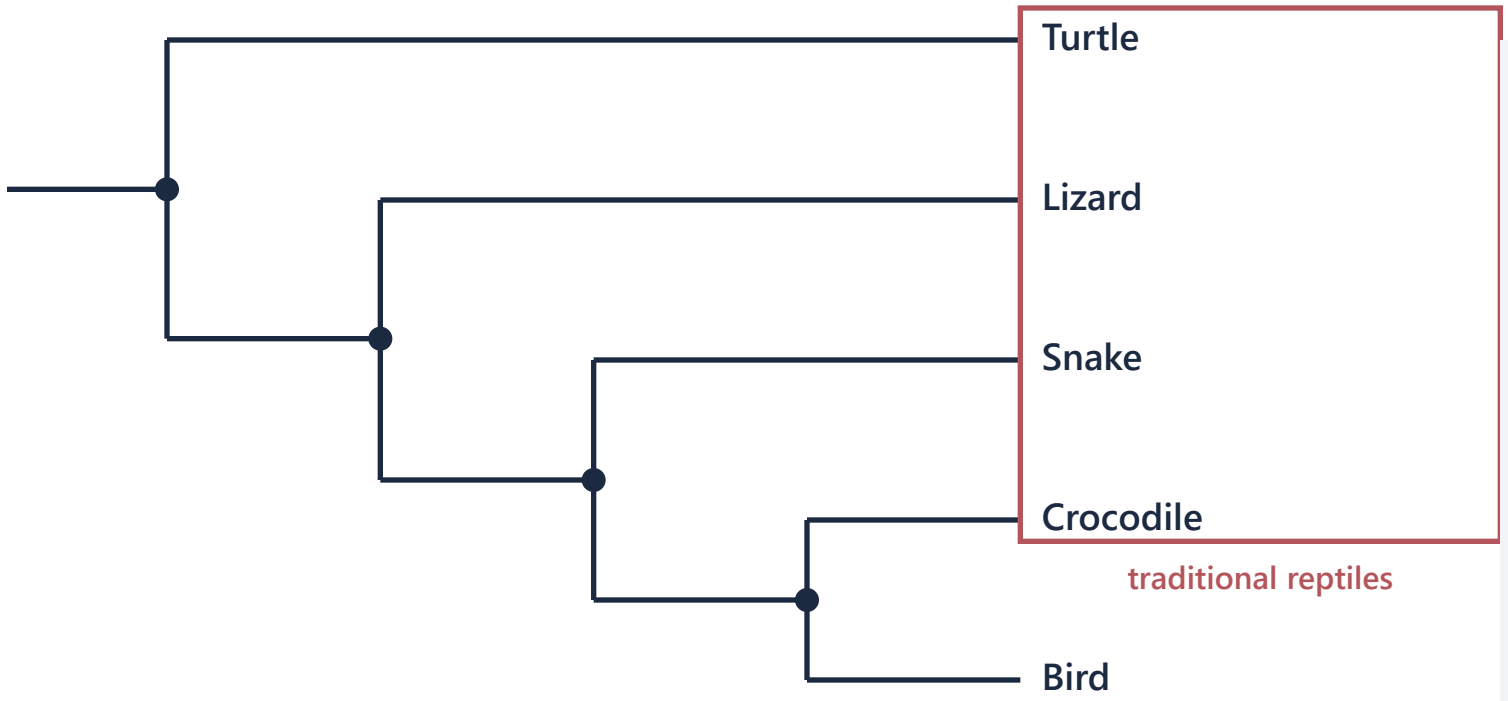
Bats fly like birds but share hair, milk and three middle-ear bones with mammals. A new bat species will make milk.

### Link

A4.1: evidence for evolution.

# Where do birds belong?

A3.2.2–3




## Read it

Crocodiles are closer to birds than to lizards. A clade including crocodiles must include birds.

# Practice: why classify?

A3.2.1–3

**List**

The eight ranks in order.

**Explain**

One problem with ranked taxa.

**Predict**

A feature of a newly found bird species.

**Use**

taxon, rank, common ancestor, convergence

# Working with data: unequal ranks

A3.2.2

| Genus | Number of species |
|---|---|
| **Homo** | 1 living |
| **Canis** | About 7 |
| **Carex (sedges)** | About 2000 |
| **Drosophila** | About 1500 |

1. Explain why a genus is not a fixed unit.
2. Suggest why sedges have so many species.
3. What would cladistics use instead?

# Check-in: class 1

A3.2.1–3

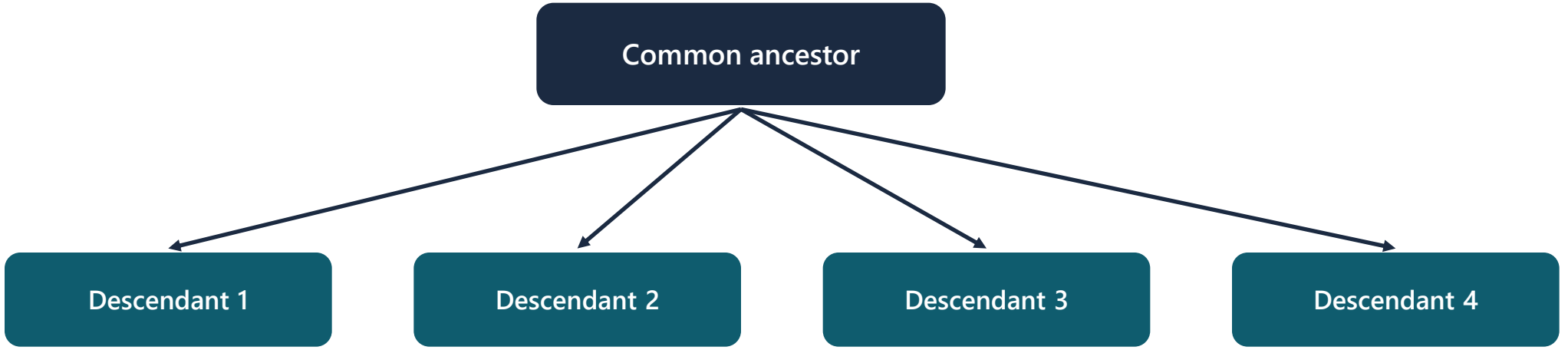
1. Give two reasons for classifying organisms.
2. List the ranks from domain to species.
3. Explain one problem with ranked taxa.
4. State the advantage of classifying by ancestry.
5. Explain why birds are a problem for traditional classification.

**How to use this**

Answer without your notes, then check against slides 3–9.

# What is a clade?

A3.2.4

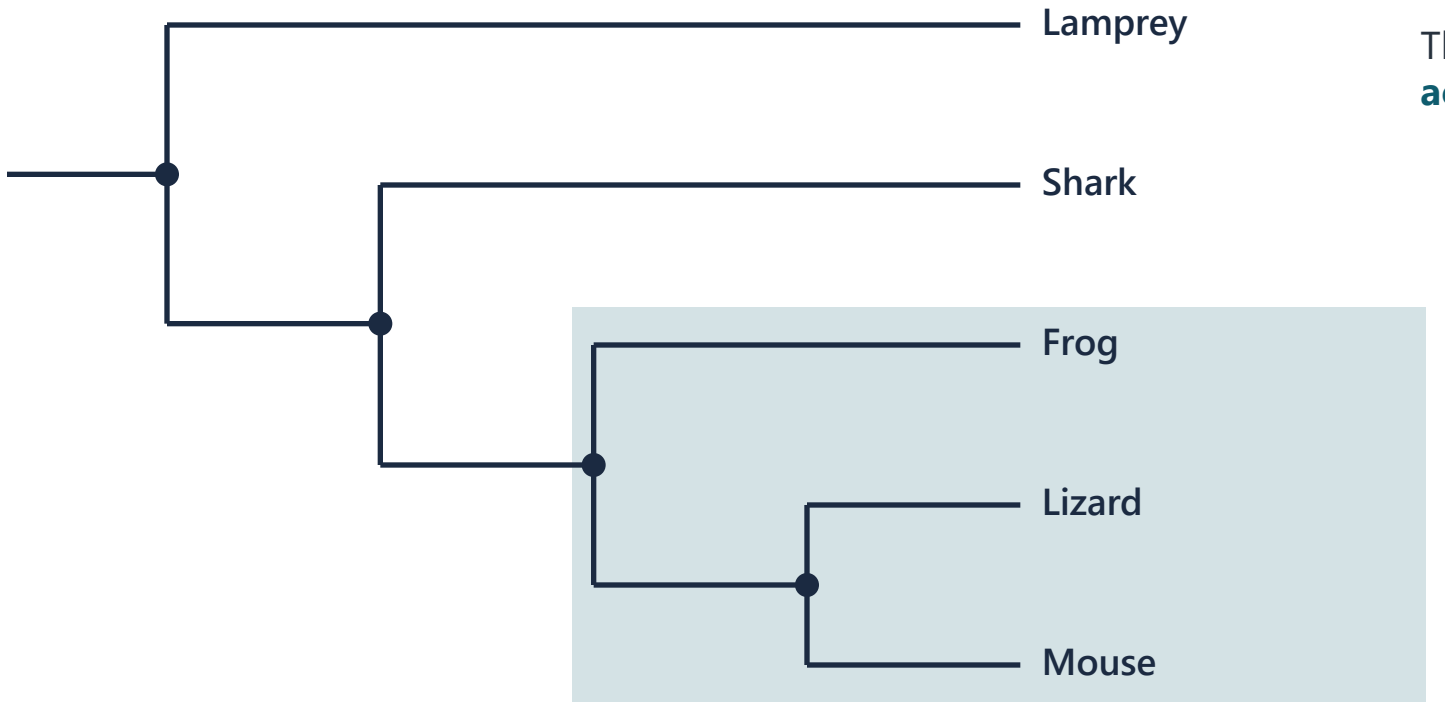



**Rule**

A clade is one ancestor and **all** of its descendants. Leave one out and it is not a clade.

# Clades on a cladogram

A3.2.4




Shaded: one clade, a common ancestor and all its descendants.

Members share inherited features.

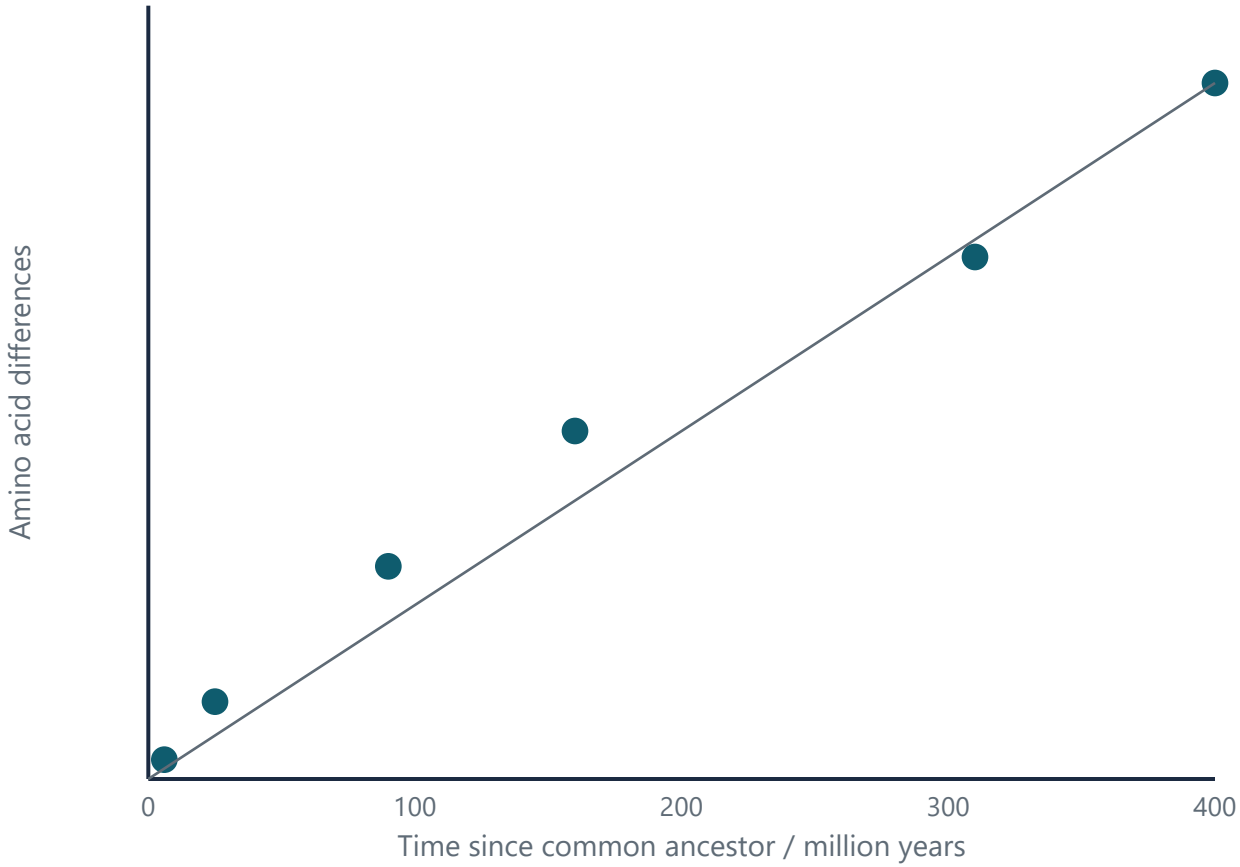
The best evidence is **base sequences** and **amino acid sequences**; morphology is used with care.

**Try it**

Is "Shark, Frog and Lizard" a clade?

# The molecular clock






Mutations build up at a roughly **steady rate**, so differences estimate time since divergence.

Calibrated with dated fossils.

### Only an estimate

Rates vary with generation time, population size, selection and gene.

# Building a cladogram from sequences

A3.2.6

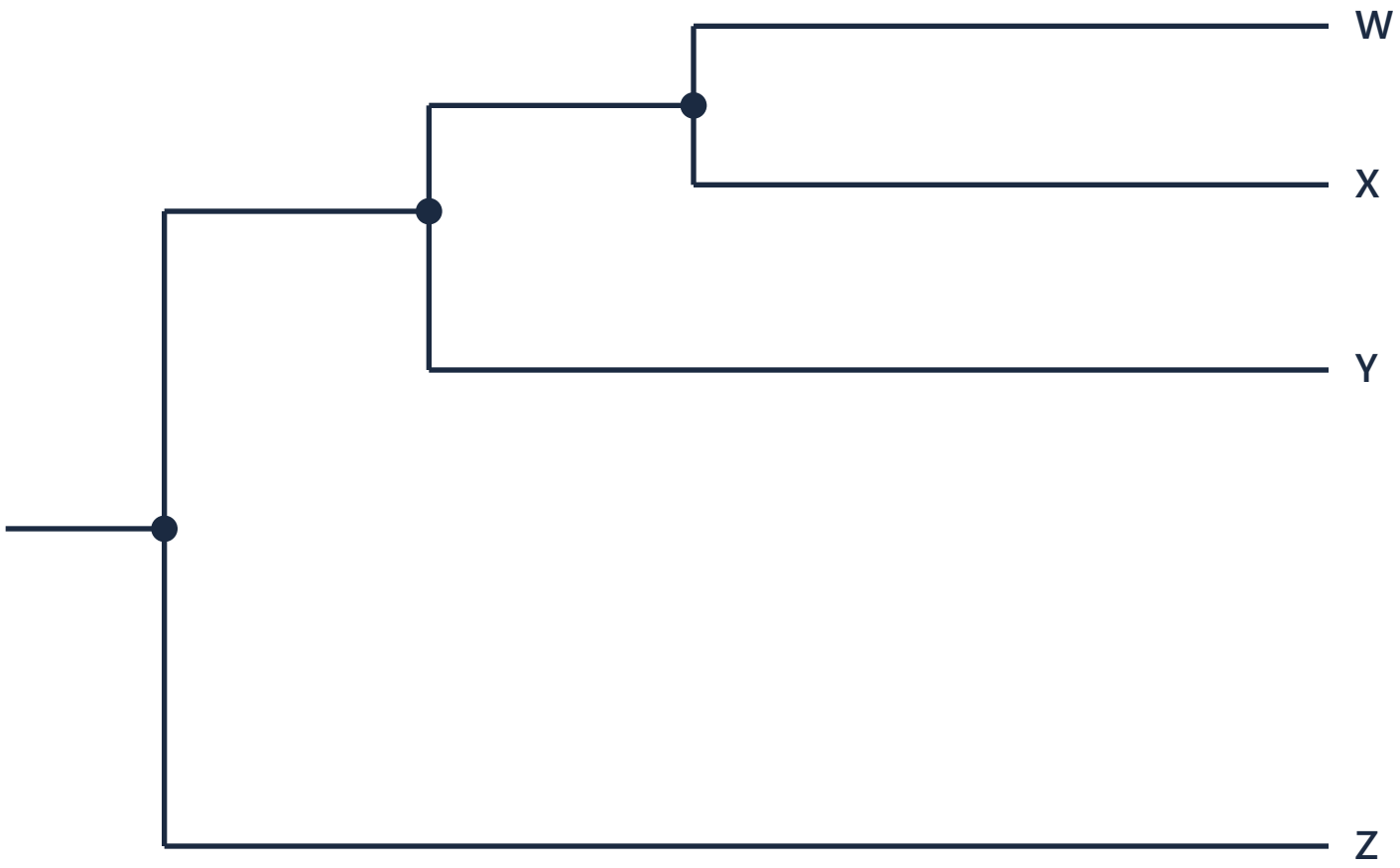
| Species | Base sequence |
|---|---|
| W | ATGCCTAGGTCA |
| X | ATGCCTAGCTCA |
| Y | ATGACTAGCTGA |
| Z | TTGACAAGCAGA |

| | W | X | Y | Z |
|---|---|---|---|---|
| W | – | 1 | 3 | 6 |
| X | | – | 2 | 5 |
| Y | | | – | 3 |

Count pair differences → **Join closest pair** → Add next closest → Add the outlier

# The result

A3.2.6

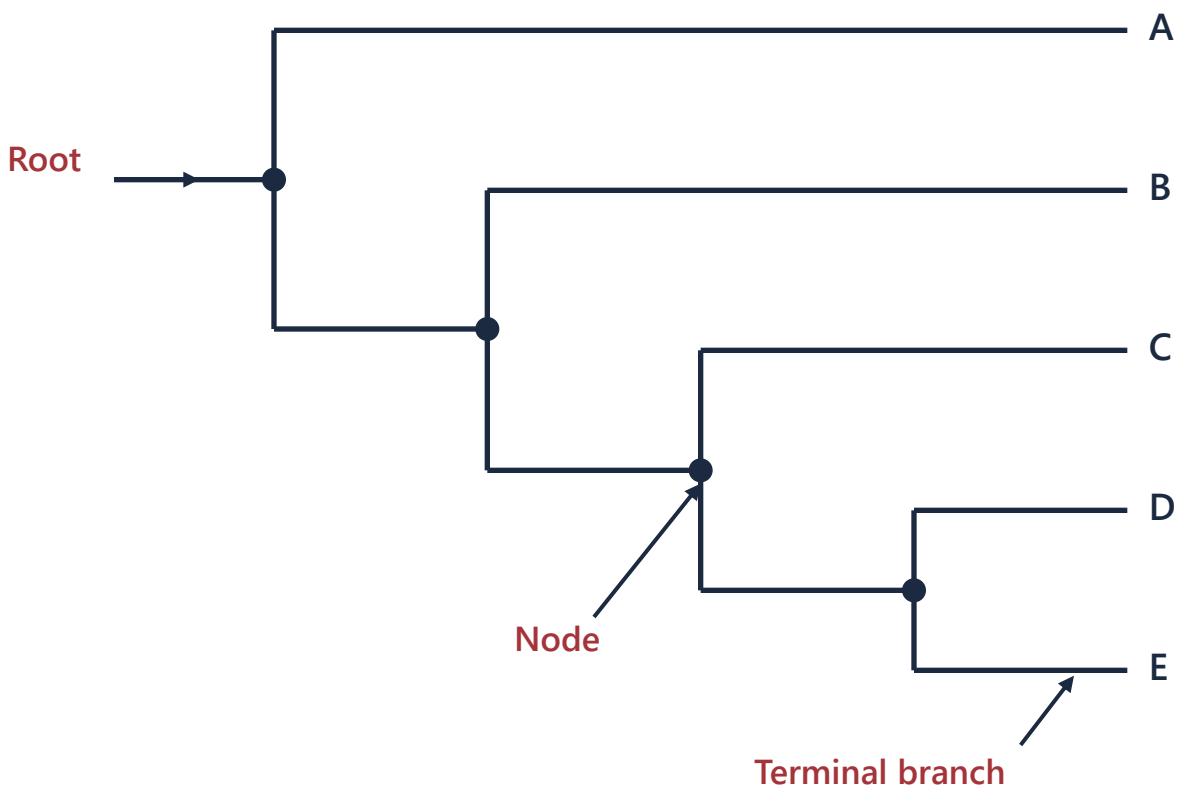



**Parsimony**

Choose the tree needing the fewest changes: each mutation is unlikely.

# Reading a cladogram

A3.2.7




**Root**: shared ancestor of all.

**Node**: hypothetical common ancestor.

**Terminal branch**: one taxon.

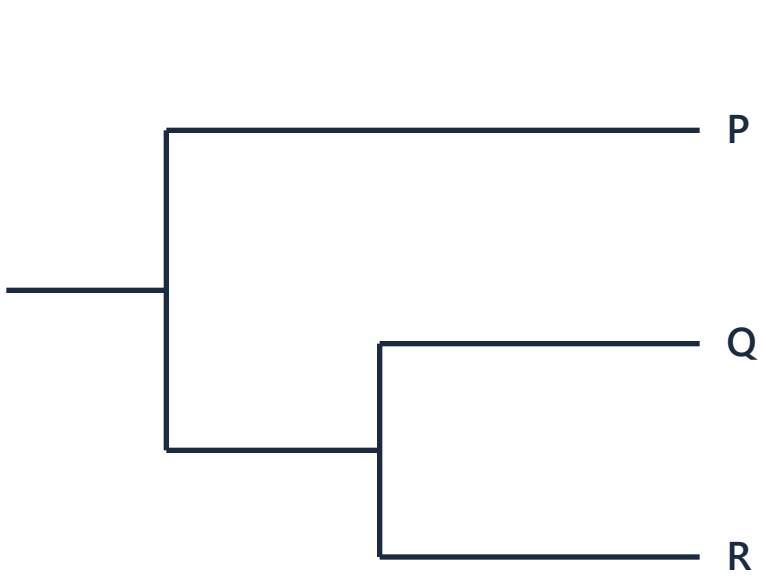
A more recent shared node = closer relatives.

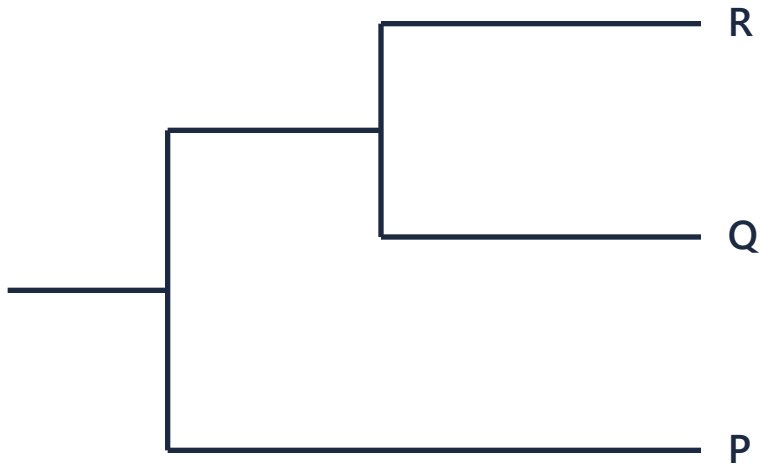
**Try it**

Smallest clade containing B and D? How many clades include C?

# Rotating branches changes nothing

A3.2.7




**Same tree**

Q and R share the most recent node in both. Order of tips means nothing; branch points do.

# Working with data: building a cladogram

A3.2.6–7

Base differences in a short DNA sequence between four species. Invented values for practice.

| | W | X | Y |
|---|---|---|---|
| **X** | 2 | | |
| **Y** | 7 | 6 | |
| **Z** | 12 | 11 | 10 |

1. Which two are most closely related?
2. Draw the cladogram.
3. Estimate when W and X split if 1 difference = 5 million years.

# Check-in: class 2

A3.2.4–7

1. Define a clade.
2. State the best evidence for clades.
3. Explain how a molecular clock works.
4. Give one reason it is only an estimate.
5. Define root, node and terminal branch.

**How to use this**

Answer without your notes, then check against slides 11–18.

# Testing a classification: the figworts

A3.2.8



Figwort, Scrophularia nodosa. H. Zell, CC BY-SA 3.0.



Greater plantain, Plantago major. Public domain.

The figwort family was based on **flower shape**.

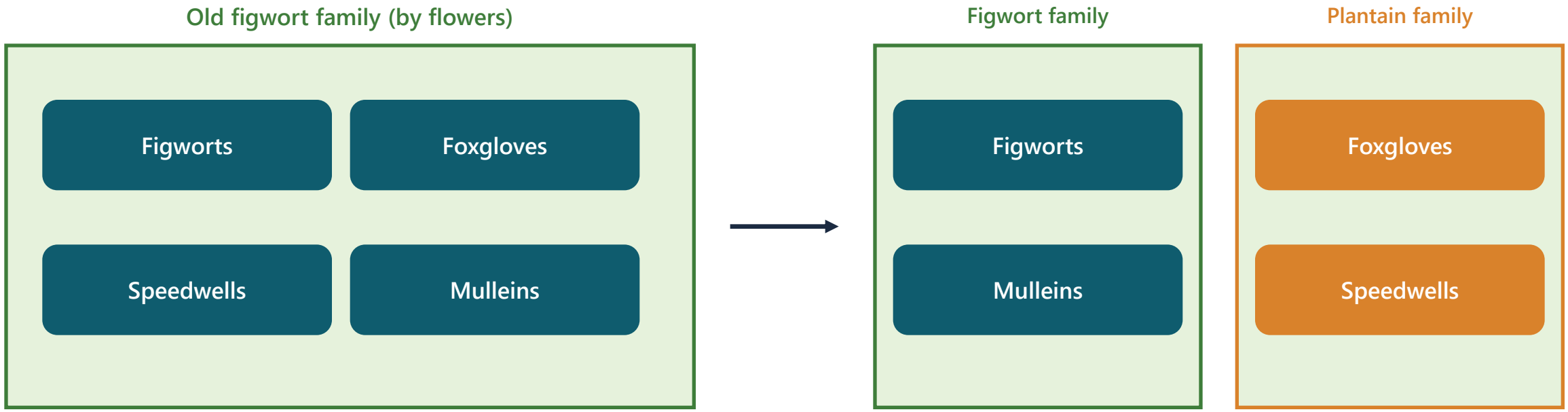
DNA showed it was **not a clade**.

**Result**

Many genera, including foxgloves and speedwells, moved to the plantain family; the figwort family lost over half its species.

# Before and after DNA

A3.2.8

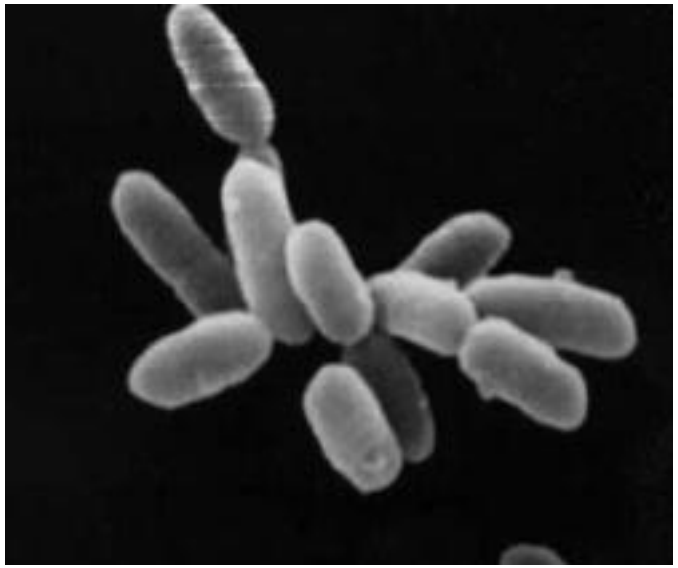



**Key idea**

When a traditional group is not a clade, the classification is changed so that groups match evolutionary relationships.

# Three domains

A3.2.9



Halobacterium, an archaean from salty water. NASA, public domain.

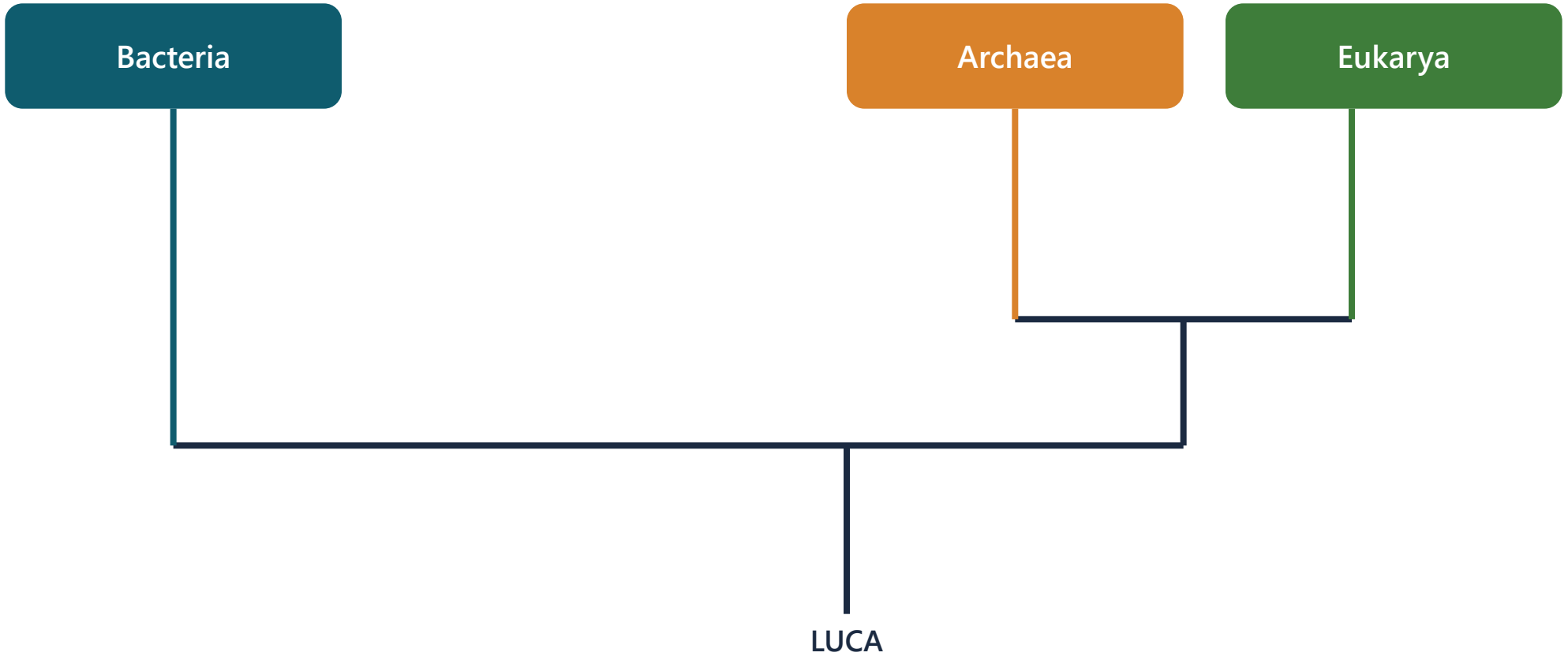
| | Bacteria | Archaea | Eukarya |
|---|---|---|---|
| **Nucleus** | No | No | Yes |
| **Histones** | No | Some | Yes |
| **Introns** | Rare | Some | Common |
| **Membrane lipids** | Ester-linked | Ether-linked | Ester-linked |

## The evidence

In 1977, rRNA sequences showed prokaryotes form two groups as different from each other as from eukaryotes. The domain rank was added: Bacteria, Archaea, Eukarya.

# The tree of life

A3.2.9




**Notice**

Archaea are closer relatives of eukaryotes than bacteria are.

# Practice: reclassification and domains

A3.2.8–9

**Explain**

Why the figwort family was split.

**Compare**

Archaea and bacteria: two differences.

**Evaluate**

Using rRNA to classify all life.

**Use**

clade, reclassify, domain, rRNA

# Check-in: class 3

A3.2.8–9

1. Explain how cladistics tested the figwort family.
2. State what happened to foxgloves.
3. Name the three domains.
4. Give two differences between archaea and bacteria.
5. State the evidence for three domains.

**How to use this**

Answer without your notes, then check against slides 20–24.

# Common mistakes to avoid

| Not this | But this |
| --- | --- |
| **Tip species evolved from each other.** | They share ancestors at nodes. |
| **Left-hand taxa are more primitive.** | Tip order means nothing. |
| **Fungi are plants.** | Fungi are a separate kingdom, closer to animals. |
| **Archaea are a kind of bacteria.** | They are a separate domain. |

# Summary: classification and cladistics

A3.2.1–9

| | Key idea |
|---|---|
| **A3.2.1–2** | Why classify; ranked taxa and their problems |
| **A3.2.3** | Classify by common ancestry |
| **A3.2.4** | Clades: ancestor + all descendants |
| **A3.2.5** | Molecular clock |
| **A3.2.6–7** | Building and reading cladograms |
| **A3.2.8** | Figworts reclassified |
| **A3.2.9** | Three domains from rRNA |

# Check that you can...

1. **Outline** the hierarchy of taxa and its problems
2. **Explain** classification by ancestry
3. **Define** a clade; **outline** the molecular clock
4. **Construct** and **interpret** cladograms
5. **Evaluate** reclassification
6. **Outline** the three domains

# Exit activity: 5 minutes

**On your own and without notes. Hand this in as you leave.**

1. Define clade.
2. State why the molecular clock gives only an estimate.
3. Outline the evidence for the three-domain system.

# Image credits

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Plantago major.jpg – Iorsh at English Wikipedia – Public domain
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